Junghyun Kim

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Postdoctoral Research Fellow, AIIS,
Seoul National University
Research Fellow, OpenMind

Research Interests

Physical AI researcher developing vision-language-action models and semantically grounded representations for generalist robots.

Education

Seoul National University — Seoul, Korea
Ph.D. in Artificial Intelligence, advised by Prof. Byoung-Tak Zhang — Mar 2021–Aug 2026

- GPA: 4.27/4.3
- **Dissertation:** Learning Semantically Grounded Latent Spaces in Vision-Language-Action Models
- Main Courses: DL, ML, AI, Pattern Recognition, RL, LLM, Cognitive Science, Neuroscience, Data Science
- Research Paper Competition: Honorable Mention (2024, 2025)

Yonsei University — Seoul, Korea
B.S. in Electrical and Electronic Engineering — Mar 2016–Feb 2021

- GPA: 4.13/4.5
- Main Courses: AI, Programming (C, Python, Matlab), Control Engineering, Mathematics, Probability, Data Structure and Algorithms, Signals, Circuits, OS, Communication Theory, Wireless and Mobile Networks
- **Thesis:** Large-Scale Tiny Face Detection using Multi-Task Cascaded Convolutional Neural Networks
- Yonsei Evison Award (2016), 1 highest honor (2020), 1 high honor (2020), 4 honors (2016, 2017, 2019)

Experience

Artificial Intelligence Institute (AIIS), Seoul National University
Postdoctoral Research Fellow — Sep 2026 - Present

OpenMind
Research Fellow — Dec 2025 - Present

Hyundai Motor Company
Project Researcher — Nov 2025 - Sep 2026

Tommoro Robotics
Research Collaborator — Jun 2024 - May 2025

BioIntelligence Laboratory, Seoul National University
Lab Manager — Dec 2023 - Jan 2026

Samsung AI EXPERT Program
Teaching Assistant (Machine Learning & Python Programming) — Jun 2023

SurroMind Corp.
Research Engineer (Developing Large-scale Visual Reasoning Model) — Aug - Dec 2022

Samsung AI Research Center, Samsung Advanced Institute of Technology (SAIT)
Student Researcher (SEM Depth Prediction via Auto Encoder) Apr - Jun 2022

Computer Science, Seoul National University
Teaching Assistant (Artificial Intelligence) Mar - Aug 2022

Robotics and Mobile Networks Lab, Yonsei University
Undergraduate Intern (Advised by Prof. Seoung-Lyun Kim) Summer 2019

Publications

** denotes equal contribution*

International Conference

[10] **Disentangling Spurious Correlations in Vision-Language-Action Models via Predicting Domain-Invariant Latent Lookahead**
J. Kim*, N. Kim*, C. Lee, S. Lee, W. Baek, A. Zhou, C. Huyen, J. K. Lee, G. C. Kang, B. T. Zhang
Conference on Robot Learning (CoRL 2026)
Semantics for Robotics Workshop @ Robotics: Science and Systems (RSS 2026) **Spotlight**

[9] **PHASOR: Phase-Anchored Universal Action Representations for Humanoid Embodiments**
K. Kim, C. Kim, J. Shin, T. Kwon, **J. Kim**, M. Koo, H. Park
Human-Motion World Models Workshop @ European Conference on Computer Vision (ECCV 2026)

[8] **MicroVLA: Edge-Deployable Vision Language Action at 10M Parameters**
N. Kim, **J. Kim**, G. C. Kang, Y. Yu, B. T. Zhang
Learning to Program Robots Workshop @ Robotics: Science and Systems (RSS 2026)

[7] **Can Embodied Agents Remember What You Said? Evaluating Dialogue-Grounded Embodied Memory in 3D Environments**
S. Jeong, **J. Kim**, G. C. Kang, B. T. Zhang
Workshop on 3D-LLM/VLA @ Computer Vision and Pattern Recognition (CVPR 2026)

[6] **EgoExo-Con: Exploring View-Invariant Video Temporal Understanding**
M. Jung, J. Xiao, **J. Kim**, B. T. Zhang, A. Yao
European Conference on Computer Vision (ECCV 2026)
Workshop on Multimodal Algorithmic Reasoning @ Neural Information Processing Systems (NeurIPS 2025)

[5] **CLIP-RT: Learning Language-Conditioned Robotic Policies from Natural Language Supervision**
G. C. Kang*, **J. Kim***, K. Shim, J. K. Lee, B. T. Zhang
Robotics: Science and Systems (RSS 2025)
3rd Workshop on Language and Robot Learning @ The Conference on Robot Learning (CoRL 2024)

[4] **Socratic Planner: Inquiry-Based Zero-Shot Planning for Embodied Instruction Following**
S. Shin, S. Jeon*, **J. Kim***, G. C. Kang*, B. T. Zhang.
IEEE International Conference on Robotics and Automation (ICRA 2025)

[3] **PGA: Personalizing Grasping Agents with Single Human-Robot Interaction**
J. Kim, G. C. Kang*, J. Kim*, S. Yang, M. Jung, B. T. Zhang.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024)

[2] **PROGrasp: Pragmatic Human-Robot Communication for Object Grasping**
G. C. Kang, **J. Kim**, J. Kim, B. T. Zhang.
IEEE International Conference on Robotics and Automation (ICRA 2024)

[1] **GVCCI: Lifelong Learning of Visual Grounding for Language-Guided Robotic Manipulation**
J. Kim, G. C. Kang*, J. Kim*, S. Shin, B. T. Zhang.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023) **Oral Presentation**

Domestic Journal

[1] **Tackling Entity Size-Based Bias in Visual Grounding**
J. Kim, G. C. Kang, H. Sul, B. T. Zhang.
KIISE Transactions on Computing Practices (KTCP 2023)

Domestic Conference

[9] **Learning Language-Conditioned Robotic Policies with Fast and Lightweight Architectures**
J. Kim, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KCC 2025, Best Paper Award)

[8] **Generalizing Language-Conditioned Robotic Policies via Large Language Models**
J. Kim, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KCC 2025, Best Paper Award)

[7] **Stochastic Trajectory Diversification for Language-Conditioned Robotic Policy Learning**
J. Kim, G. C. Kang, K. Shim, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KSC 2024, Best Paper Award)

[6] **Data Collection Method for Robotic Action Learning through Natural Language Commands**
J. Kim, G. C. Kang, K. Shim, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KSC 2024)

[5] **Personalizing Large Language Models via Interest Trait Summarization**
J. Kim, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KSC 2023)

[4] **Unsupervised Adaptation for Zero-shot Visual Grounding via Pseudo Query Generation**
J. Kim, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KCC 2023)

[3] **Referring Expression Segmentation on Small Objects**
J. Kim, G. C. Kang, H. Sul, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KSC 2022, Best Paper Award)

[2] **Attention-based Text Augmentation Method for Referring Expression Segmentation**
J. Kim, G. C. Kang, H. Sul, B. T. Zhang.
Korean Institute of Information Scientists and Engineers (KCC 2022)

[1] **Adaptive Spatial Comprehension via Object Relationship Learning with Home Robot**
J. Kim, S. Lee, Y. Yoo, B. T. Zhang
Korean Institute of Information Scientists and Engineers (KSC 2021)

Patents

- (US, KR) Robot control system for performing contrastive imitation learning method for robot and contrastive imitation learning method for robot using the same
- (PCT, KR) Method for Generating Artificial Intelligence Model Using Natural Language Instruction, Method and Apparatus for Driving Robot Using Said Artificial Intelligence Model
- (PCT, KR) Method and Apparatus for Generating Natural Language Instruction Based on Visual Recognition
- (PCT, KR) Method And Apparatus For Creating An Artificial Intelligence Model That Recognizes Personalized Objects, And Robot Control Method And System Using The Same
- (KR) Method and Device for Interactive Object Grasping with Pragmatic Inference

Research Projects

Large-Parameter Action Foundation Models for Embodied Intelligent Agents Solving Real-life Problems of Real-World in Real-time
National Research Foundation of Korea (NRF) 2024–Present

Development of humanoid robots that feel like humans, communicate, and learn
Korea Evaluation Institute of Industrial Technology (KEIT) 2024–Present

Embodied Artificial Intelligence Interacting with Humans in Real World
National Research Foundation of Korea (NRF) 2023–Present

Self-directed AI Agents with Problem-solving Capability
Institute for Information Communications Technology Planning Evaluation (IITP) 2022–Present

Development of Uncertainty-Aware Agents Learning by Asking Questions
Institute for Information Communications Technology Planning Evaluation (IITP) 2022–Present

Professional Activities

- **(Student Member)** IEEE Robotics and Automation Society 2025
- **(Reviewer)** IEEE International Conference on Robotics & Automation (ICRA) 2026
- **(Reviewer)** Conference on Robot Learning (CoRL) 2026
- **(Reviewer)** RSS 2026 Workshop from Perception to Action
- **(Reviewer)** IEEE Robotics and Automation Letters (RA-L) 2024
- **(Reviewer)** IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024
- **(Mentor)** POSCO DX AI Youth Challenge 2022-2026
- **(Member)** Korean Institute of Information Scientists and Engineers (KIISE) 2021-2025

Skills

- **Programming:** Python, C, Matlab, Verilog
- **Robots:** UR5, Kinova Gen3, TurtleBot4, HSR

Languages

- **Korean:** Native
- **English:** Fluent (New TEPS: 483)

Extracurricular Activities

- Seoul National University President's Cup 2025
  Amature Golfer / Runner-up in the finals
- Professional Golf Tour in Asia (Asian Tour) 2022
  Professional Golf Caddie
- Korea Professional Golfers' Association (KPGA) 2021-2022
  Professional Golf Caddie
- Seoul Junior Athletics Championships, Korean Olympic Committee (KOC) 2014
  Junior Athlete / 2^{nd} place in Seoul, Korea
- Korea Junior Golf Association 2011–2013
  Junior Golfer